

Mark Vincent Francisco

CS-499: Computer Science Capstone

Professor Joseph Conlan

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3-2 Milestone Two: Enhancement One: Software Design and Engineering

Narrative:

1. Briefly describe the artifact. What is it? When was it created?

The artifact selected for enhancement in fulfillment of category one, Software Design and Engineering outcomes was an event planning/tracking mobile application for Android that was developed in a CS-360: Mobile Architect and Programming course back in March, 2026. The event planning/tracking app was designed to provide users a tool in which to store upcoming events within a database (i.e. SQLite) along with useful information along with the event such as: the title, description, date, and phone number. The information associated with each event is displayed within a RecyclerView of the user interface giving users seamless scroll functionality in order to view all events stored within the database. There is also the functionality to delete an event, or update the title, description, or date of the event by simply clicking the update button. A notification will be sent to the user's device and the phone number associated with the event when an event is within 7 days of the present day. There is also the function to create a username and password for the application and store that information within a database and use it to login along with a sign out function as well.

2. Justify the inclusion of the artifact in your ePortfolio. Why did you select this item?

What specific components of the artifact showcase your skills and abilities in software development? How was the artifact improved?

The reason why I selected this artifact to include in my ePortfolio is because it satisfies all required categories (A, B, and C) for the entire project for this course. This artifact calls for enhancements on multiple items within the code to fulfill the Software Engineering and Design category through improving input validation, error handling, software architecture, and modularity through the implementation of MVC architectural pattern. Specific components within this project such as: the username and password input fields, event information fields (i.e. phone number, date, etc.) and package architectural design were all items that showcase my skills and abilities in software development. Enhancement of these specific components of my artifact showcases my ability in defensive programming, architectural refactoring, and software engineering principles.

3. Did you meet the course outcomes you planned to meet with this enhancement in Module One? Do you have any updates to your outcome-coverage plans?

Yes, I have closely met two course outcomes that I planned to meet with this particular enhancements which are:

- a. *employing* strategies for building collaborative environments that enable diverse audiences to support organizational decision making in the field of computer science.

- b. *develop* a security mindset that anticipates adversarial exploits in software architecture and designs to expose potential vulnerabilities, mitigate design flaws, and ensure privacy and enhanced security of data and resources.

I currently do not have any updates to my outcome-coverage plans.

- 4. Reflect on the process of enhancing and modifying the artifact. What did you learn as you were creating it and improving it? What challenges did you face?

What I learned as I was making improvements to my artifact was the realization how much more was needed to make the application viable for deployment. Having completed almost all the computer science courses at Southern New Hampshire University, I have so much more context as to how a software project is supposed to look as opposed to what I thought back then as I was first creating this artifact. I also learned how difficult it is to re-acclimate to older projects if subpar comments or undesirable coding practices were previously implemented. Challenges I faced was having to research how an MVC pattern was supposed to look like, as it has been a couple of months since I looked into that pattern.